Mitosis

Exam code: 0610 & 0970

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New Cells are Made by Mitosis: Extended

Extended Tier Only

Mitosis – Basics

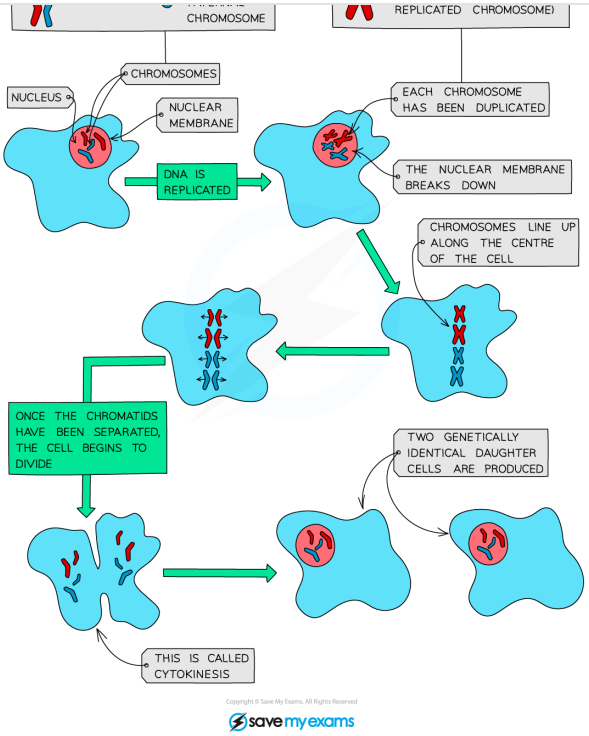
- Most body cells have two copies of each chromosome
- We describe these cells as **diploid**
- When cells divide their chromosomes double beforehand
- This ensures that when the cell splits in two, each new cell still has two copies of each chromosome (is still diploid)
- This type of cell division is used for **growth, repair of damaged tissues, replacement of cells** and **asexual reproduction** and is known as **mitosis**
- Mitosis is defined as **nuclear division giving rise to genetically identical cells**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



The process of cell division by mitosis

Process:

- Just before mitosis, each chromosome in the nucleus copies itself exactly (forms x - shaped chromosomes)
- Chromosomes line up along the centre of the cell where cell fibers pull them apart
- The cell divides into two; each new cell has a copy of each of the chromosomes

Importance:

- All cells in the body (excluding gametes) are produced by mitosis of the zygote
- Mitosis is important for replacing cells e.g, skin cells, red blood cells and for allowing growth (production of new cells e.g. when a zygote divides to form an embryo)

Occurs in:

- *Growth*: mitosis produces new cells
- *Repair*: to replace damaged or dead cells
- *Asexual reproduction*: mitosis produces offspring that are genetically identical to the parent

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Mitosis & Stem Cells: Extended

Extended Tier Only

- Many tissues in the human body contain a small number of **unspecialised cells**
- These are called **stem cells** and their function is to divide by mitosis and **produce new daughter cells that can become specialised** within the tissue and be used for different functions
- The ultimate stem cell is the **zygote**
  - A zygote divides several times by mitosis to become a ball of unspecialised cells (around 200–300 cells)
    - These are **embryonic stem cells**
  - These cells are all the same and start differentiating as the fetus develops with recognisable features

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